

Financial Analysis of Startups Addressing Air Pollution in India:

Identifying Optimal Investment Opportunities

By

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ABSTRACT

By the American Lung Association's standards, an air quality index of 301-500 is considered hazardous, which is the worst category on their chart displaying AQI values. A hazardous classification of AQI recommends that everyone within the area should avoid all physical activity outdoors.

Using Forecast models (can be applied where historical numerical data is available to show future trends), I have analyzed the trend of air quality over time and in the future and determined which startups would be beneficial to invest in (based on their projected growth) to solve the problem of air pollution in India. With this, I have created an angel investment portfolio with \$1 billion to invest in these startup companies and university laboratories that are experimenting with biodegradable removal of the main components of air pollution.

In India today, arguably the biggest problems at hand are the air quality and the air pollution, specifically in the northern states. This problem is not only deadly for current Indian citizens, but also future citizens who haven't been born yet. A pregnant woman in India who is experiencing the effects of air pollution can suffer a variety of issues such as a miscarriage, early delivery, low birth weight, and other development/physical issues for the child. Once these children are born, air pollution can harm their cognitive and physical development, causing irreversible negative effects in their health. Air pollution also causes 6.7 million premature deaths per year. This is not

just a social problem, though. Currently, deaths related to air pollution take up 3 trillion USD, which accounts for 3.3% of the world's GDP. If this issue were to be resolved, money in various countries' GDPs could be freed up, allowing for governments to work towards solving other pressing issues. In our world today, none of the world's 100 biggest cities (with Delhi being #14) pass the WHO's guidelines on air pollution, which were created in September 2021. This reflects the unsafety of air quality and creates possibility for worsening in the future.

This project and air pollution are especially important to me as a child of Indian parents who immigrated to America. As a kid, we would go to India routinely and I would experience the effects of air pollution firsthand. This would result in me disliking India trips, affecting how I treated the family I saw there and the opportunities that I had with family members that I did not get to see often. As someone who suffers from asthma, the air pollution in India would have a greater effect on me as I was forced to carry an inhaler and nebulizer around wherever we went in the case that I experienced an asthma attack.

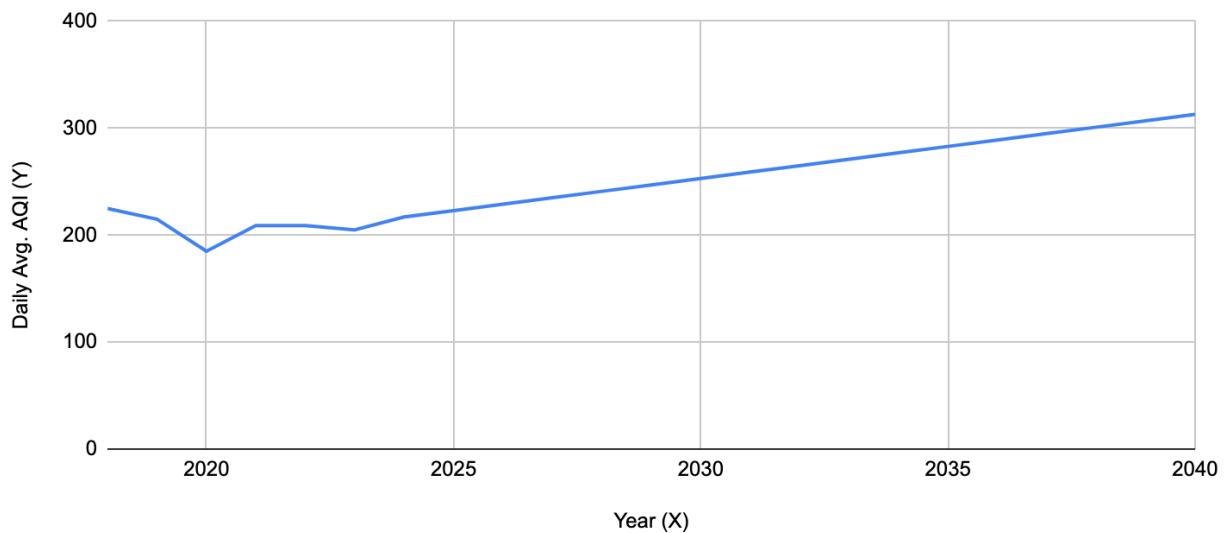
Using a forecast model, I have found that by 2040, the average daily AQI (air quality index) in Delhi will be above 300, which is considered hazardous by the American Lung Association.

To show the trend of India's daily AQI (air quality index), I gathered data from past years in Delhi and created a forecast model. I then input this data into Google Sheets by date and used the Forecast function to numerically display the predicted values for India's AQI in future years.

After gathering the predicted data for future years, I used Google Sheets to turn the numerical AQI values into a graph.

After analyzing the finished model, I found that by 2040, India's average daily AQI will reach 313. According to the American Lung Association, an AQI index above 301 is considered hazardous and is the worst classification on their chart of AQI value ranges. Based on these findings, I can confidently say that necessary measures must be taken to reverse the effects of India's steadily worsening air quality before it is too late.

Daily Avg. AQI (Y) vs. Year (X)



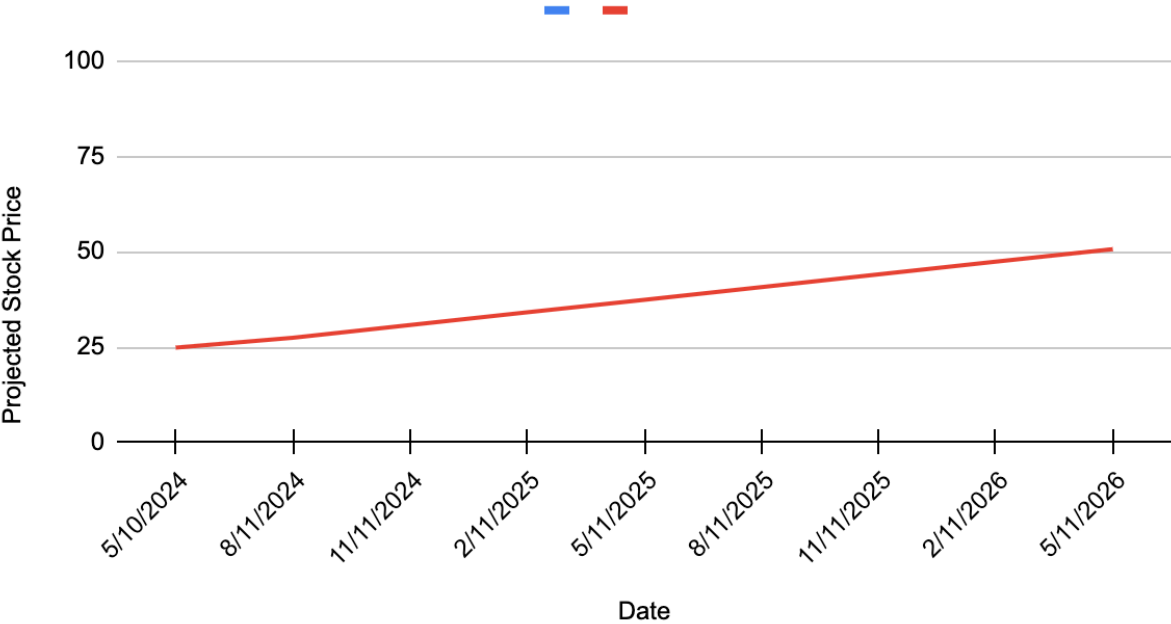
Our Investment Portfolio

- **Tetra Tech TTEK: firm focused on improving environmental services**
 - <https://www.tetratech.com/>
- **Brookfield Renewable Partners LP: focuses on renewable energy and reduced use of fossil fuels**
 - <https://bep.brookfield.com/>
- **Fuel Tech, Inc.: creates products aimed at reducing air pollutants and focuses on building an energy-efficient environment**
 - <https://www.ftek.com/>
- **CECO Environmental: develops air pollution technology and services**
 - <https://www.cecoenviro.com/>
- **Donaldson Company, Inc.: produces and markets air filters**

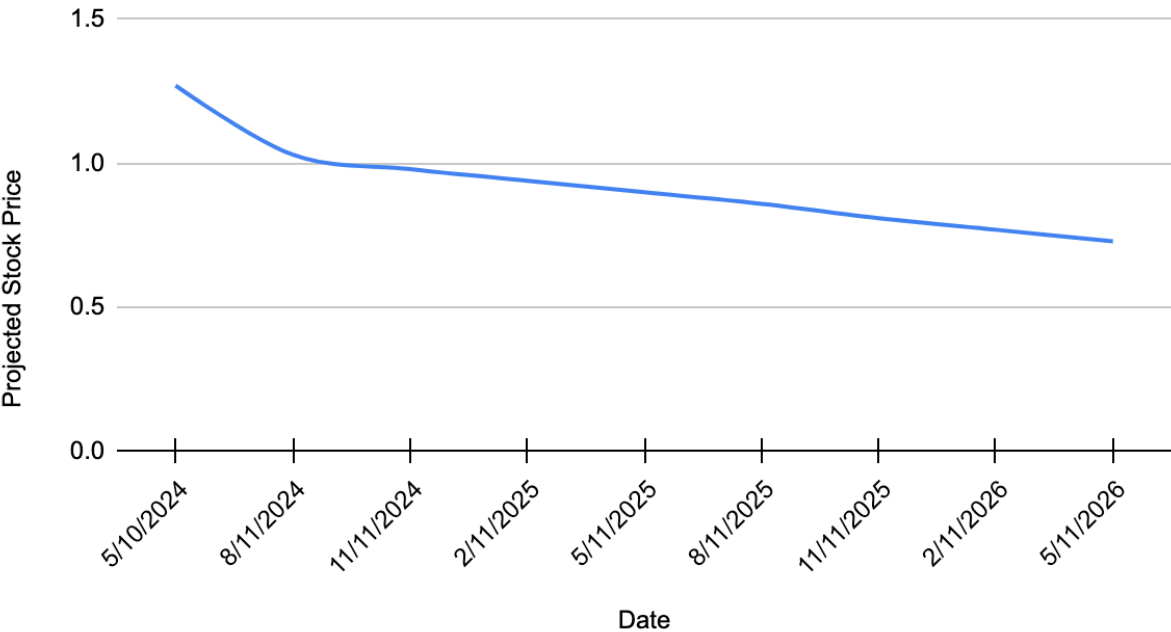
- <https://www.donaldson.com/en-us/>

After putting together a portfolio of startups that are dedicated to decarbonization and reversing the effects of air pollution, I used financial analytical skills to determine which ones would be most beneficial to invest in. For each startup, I gathered their daily stock prices for the past year and used a forecast model to determine their projected stock prices for the next 2 years. After getting this numerical data, I used Google Sheets to create a graph with a trend line displaying the company's daily stock price over the next 2 years.

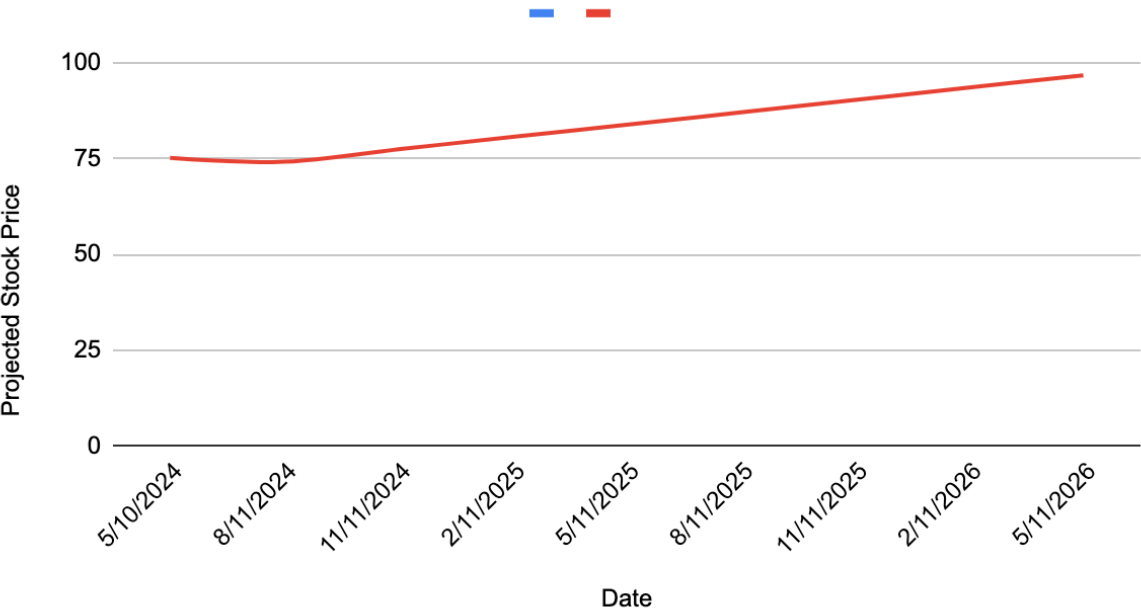
CECO Environmental Projected Stock Price vs. Date



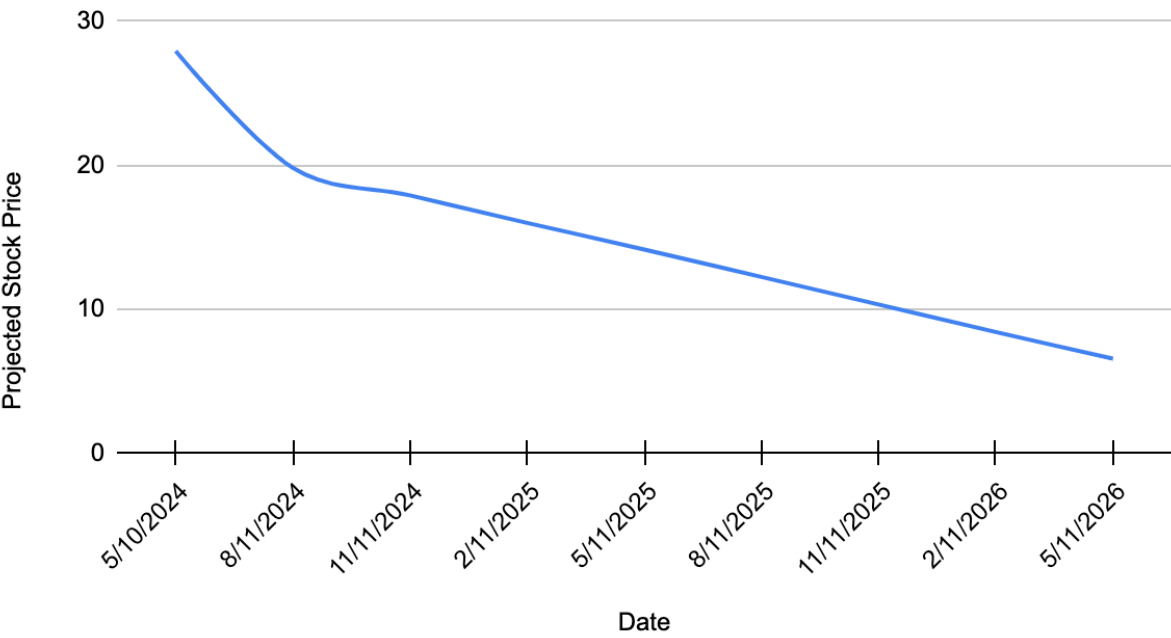
Fuel Tech, Inc. Projected Stock Price vs. Date



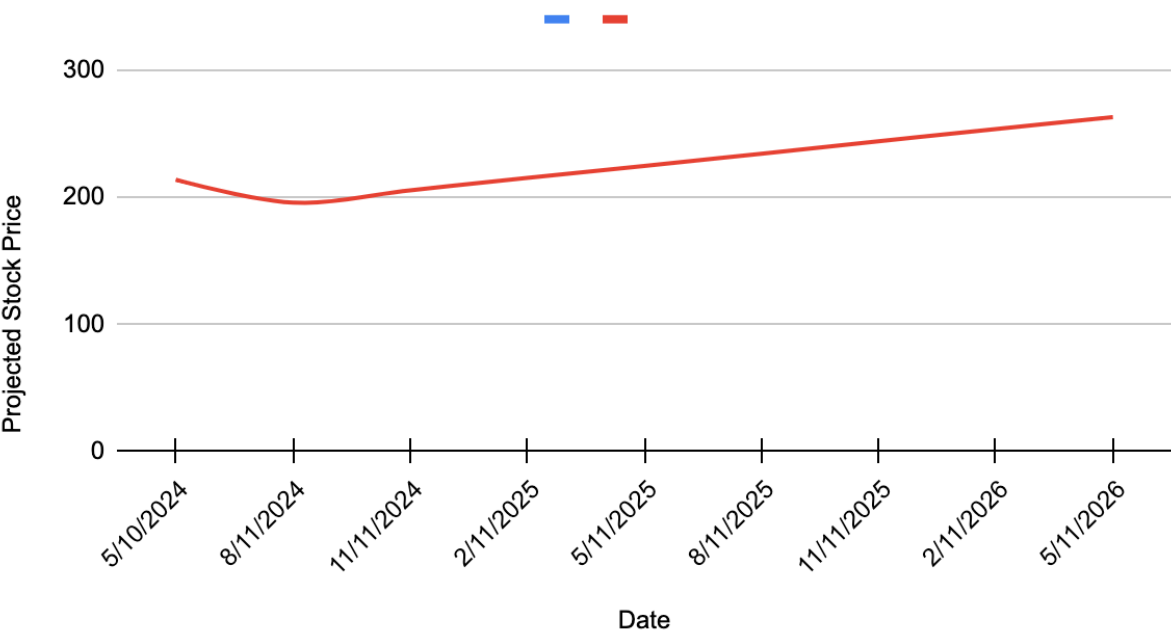
Donaldson Company, Inc. Projected Stock Price vs. Date



Brookfield Renewable Partners Projected Stock Price vs. Date



TetraTech Projected Stock Price vs. Date



How We Came to Our Answer

After creating and analyzing the models for each of the startups, I found that the most beneficial startups to invest in would be Donaldson Company, Inc. and TetraTech. In the forecast model, both of the startups' stock prices are projected to increase over the next 2 years, showing growth. Donaldson Company, Inc. also has a P/E (price/earnings) ratio of 20.66 while TetraTech has a P/E ratio of 41.70, both of which show a promising return on investment if we were to invest in these startups. Both companies also have a relatively high stock price in comparison to the other startups in my portfolio, showing investors' confidence and possible further growth.

Based on these findings, with \$1 billion in my angel investment portfolio, I would decide to split it evenly between Donaldson Company, Inc. and TetraTech as both companies show promising growth and would be effective in solving India's air quality problem. In my opinion, I believe this would not only halt the steady decline in AQI, but also reverse it and contribute to an improvement in India's air quality, especially in metropolitan areas such as Delhi.

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